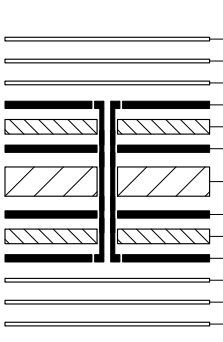
Tessera Badge Fabrication Document

Layer Stack Legend



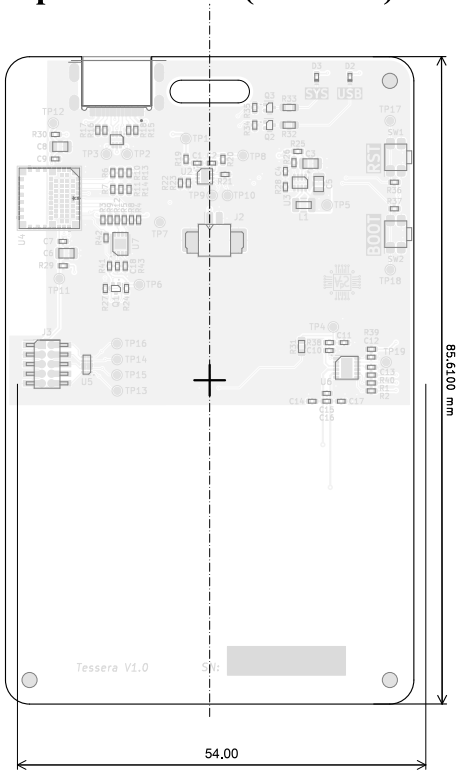
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1164mm	FR4_2116	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.0152mm (0.43oz)**		**Signal**	**GBR**
Core		0.665mm	FR4	Dielectric	
Copper	**L3 (GND)**	**0.0152mm (0.43oz)**		**Plane**	**GBR**
Prepreg		0.1164mm	FR4_2116	Dielectric	
Copper	**L4 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.0382mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	90	±10 %	L4	0.20	0.28	L3

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY APS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	54.000 × 85.610 mm
BOARD THICKNESS	1.038 mm
TRACE WIDTH	0.120 mm
TRACE TO TRACE	0.120 mm
MIN. HOLE (PTH)	0.200 mm
MIN. HOLE (NPTH)	N/A mm
ANNULAR RING	0.050 mm
COPPER TO HOLE	0.150 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

12) CONFIRM SPACE WIDTHS AND SPACINGS.

Comments:	Company: APS		Variant: PRELIMINARY	Git Hash: 4a3859d
	Board Name: **Tessera Badge**		Project Name: **Tessera**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: tessera.kicad_pcb	Designer: Antonio Morales	Date: 2026-10-01	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **8**